1. Executive Summary

The Marketing department currently faces significant operational friction in content creation. Each time a new product feature is launched or a technical blog is published, human writers must manually repurpose that content across multiple platforms — transforming long-form articles into LinkedIn posts, Twitter/X threads, and formal client newsletters.

This manual process generates two critical business problems: Creative Burnout, where repetitive rewriting of identical information exhausts creative capacity, and Content Inconsistency, where tone and messaging drift across channels, undermining brand integrity.

The Autonomous Content Factory resolves both problems by introducing an AI-powered, multi-agent assembly line. A single source document is ingested once and automatically transformed into a coordinated, brand-aligned marketing campaign across all required channels — without manual intervention.

Problem	Solution
Manual content repurposing causes creative burnout, brand inconsistency, and slow time-to-market for every product launch.	An AI-driven, three-agent pipeline autonomously generates, validates, and publishes multi-channel content from a single source document.

2. Product Overview

The Autonomous Content Factory is a web-based SaaS platform built around three specialized AI agents that collaborate in a structured pipeline. Each agent has a defined role, defined inputs, and defined outputs, ensuring a deterministic and auditable content generation process.

2.1 System Architecture Overview

Agent	Role	Input	Output
Agent 1 — Researcher	Analytical Brain: Extracts truth from raw source material	Raw document / URL	Structured Fact-Sheet (JSON/MD)

Agent	Role	Input	Output
Agent 2 — Copywriter	Creative Voice: Transforms facts into multi-format content	Fact-Sheet	Blog, Social Thread, Email Teaser
Agent 3 — Editor	Gatekeeper: Audits and approves all content	Drafts + Fact-Sheet	Approved content or Correction Notes

3. User Personas

Persona	Goals	Pain Points
Content Marketer	Rapidly repurpose launches and blogs across channels	Repetitive writing; inconsistent tone
Marketing Manager	Ensure brand consistency; review final outputs quickly	Lack of visibility into content pipeline
Product Manager	Translate technical docs to marketing copy	Disconnect between engineering and marketing teams

4. Detailed Feature Requirements

Feature 1: Lead Research & Fact-Check Agent ("The Analytical Brain")

This agent serves as the foundation of the entire pipeline. Its sole responsibility is to read raw source material and produce a reliable, structured "Source of Truth" that all downstream agents are bound to follow.

Feature 1 Requirements	
1	Extraction: Identify the core product features, technical specifications, and target audience from a raw text file or URL input.
2	Fact-Sheet Creation: Produce a structured Source of Truth document in JSON or Markdown format. This document is the authoritative reference for all other agents.
3	Value Proposition Identification: Explicitly identify and label the primary value proposition within the Fact-Sheet so the Copywriter Agent can center every piece of content around it.

4	Ambiguity Flagging: Detect and flag any ambiguous or unclear statements in the source text that could cause downstream misinterpretation or hallucination.
5	Source Traceability: Every claim in the Fact-Sheet must include a reference to its location in the source document (line number or paragraph ID).

Feature 2: Creative Copywriter Agent ("The Voice")

This agent receives the validated Fact-Sheet from Agent 1 and transforms it into three distinct content formats simultaneously. It must adapt its tone for each channel while keeping the core value proposition central to every piece.

Feature 2 Requirements	
1	Multi-Format Generation: Simultaneously produce (a) a 500-word Blog Post with introduction, body, and conclusion; (b) a 5-post Social Media Thread optimized for Twitter/X; and (c) a 1-paragraph Email Teaser for client newsletters.
2	Tone Control — Professional/Trustworthy: Blog posts and email teasers must use a formal, authoritative tone aligned with brand guidelines.
3	Tone Control — Engaging/Punchy: Social media threads must use active voice, short sentences, and high-energy language appropriate for the platform.
4	Value Proposition Centering: The specific value proposition identified by Agent 1 must serve as the narrative anchor in all three content formats.
5	Instruction Compliance: The agent must adhere strictly to the Fact-Sheet and must not introduce information, features, or claims not present in the source document.
6	Parallel Processing: All three formats must be generated in a single pass to maximize throughput and minimize latency.

Feature 3: Editor-in-Chief Agent ("The Gatekeeper")

This agent does not generate content. Its function is exclusively critical review and quality assurance. It compares every draft produced by Agent 2 against the Fact-Sheet and brand standards, and it either approves the content or sends it back with specific, actionable correction notes.

Feature 3 Requirements	
1	Hallucination Check: Perform a claim-by-claim comparison between each content draft and the Fact-Sheet. Any invented feature, incorrect price, or unverifiable claim triggers an automatic rejection.

2	Tone Audit: Evaluate each draft for tone appropriateness. Reject content that is too salesy, too robotic, or inconsistent with the intended channel tone.
3	Brand Consistency Review: Verify that the value proposition is prominently featured and that brand voice guidelines are maintained throughout.
4	Structured Feedback Loop: When a draft is rejected, generate a Correction Note that (a) identifies the specific issue, (b) references the exact location in the draft, and (c) provides an actionable instruction for the Copywriter Agent (e.g., "Blog post paragraph 2: removes claimed price of $99; the Fact-Sheet lists no pricing information").
5	Approval Gating: Content may not proceed to the Campaign Assembly Dashboard unless it has been explicitly approved by this agent.
6	Maximum Revision Cycles: Enforce a maximum of three revision cycles per content piece before escalating to a human reviewer.

Feature 4: Campaign Assembly Dashboard

The Campaign Assembly Dashboard is the user-facing interface that surfaces the agent collaboration in real-time and presents final approved content for review, editing, and export.

Feature 4 Requirements	
1	Live Agent Activity Feed: A scrolling visual log displaying agent actions in real time (e.g., "[10:42:03] Editor Agent rejected LinkedIn post — returning to Copywriter for revision 2/3..."). Each log entry must display the agent name, timestamp, action type, and content piece affected.
2	Agent Status Indicators: Visual icons representing each of the three agents with real-time states: Idle, Thinking, Typing, Reviewing, and Approved.
3	Side-by-Side Comparison View: Display the original source document alongside all three final approved content outputs in a split-panel layout.
4	One-Click Campaign Export: A button that packages the Blog Post, Social Thread, Email Teaser, and the original Fact-Sheet into a single downloadable ZIP archive.
5	Clipboard Copy: Per-channel copy-to-clipboard buttons for quick pasting into external platforms.
6	Responsive Preview Toggle: A view switcher that renders the Social Thread in a mobile-simulated screen (375px width) and the Blog Post in a desktop-simulated view (1280px width).

5. Additional Platform Features

5.1 Authentication — Login & Sign-Up

A secure authentication system governs all access to the platform, protecting proprietary source documents and generated campaign content.

- Email / password registration with email verification flow
- Social sign-in via Google OAuth 2.0 and Microsoft Azure AD
- JWT-based session management with configurable expiry (default: 8 hours)
- Password reset via time-limited email link (expires in 24 hours)
- Role-based access control: Admin, Manager, and Contributor permission tiers
- Multi-factor authentication (MFA) optional for standard users, mandatory for Admin role

5.2 AI Correction Chatbot

An embedded conversational assistant allows users to request targeted revisions to any approved content piece without re-running the full agent pipeline.

- Persistent chat panel accessible from the Campaign Assembly Dashboard for each content piece
- Natural language correction interface: Users describe desired changes in plain English (e.g., "Make the blog introduction shorter and remove the third bullet point")
- Context-aware responses: The chatbot has access to both the current draft and the original Fact-Sheet, ensuring corrections remain factually grounded
- Revision history within session: Users can undo chatbot-applied changes and compare versions
- Escalation path: Chatbot corrections that would violate the Fact-Sheet are flagged and require user confirmation before application

5.3 Project History Tracker

A persistent history system records every campaign generation run, enabling teams to audit past work, retrieve approved content, and track performance trends over time.

- Project List View: Chronologically sorted list of all past campaign runs, displaying project name, source document, date created, and current status (Draft / Approved / Exported)
- Full State Restore: Clicking any historical project reopens the Campaign Assembly Dashboard in its final approved state, including the agent activity log
- Search and Filter: Filter by date range, content status, or keyword search across project names and content
- Soft Delete with 30-Day Retention: Deleted projects enter a trash state and are permanently purged after 30 days; recoverable within the window
- Export History Log: Download a CSV audit trail of all project activity including agent revision counts, approval times, and export events

6. End-to-End User Flow

Step	Screen / Module	User Action & System Response
1	Login / Sign-Up	User authenticates via email+password or social sign-in. System validates credentials and redirects to the Dashboard.
2	Campaign Start Page	User uploads a technical document (PDF/DOCX/TXT) or pastes a URL. System validates file format and initiates the pipeline.
3	Agent Room — Research	Agent 1 processes the source document. User sees a "Thinking" state animation. Agent 1 outputs the Fact-Sheet.
4	Agent Room — Copywriting	Agent 2 receives the Fact-Sheet and generates all three content formats. User sees a "Typing" state animation.
5	Agent Room — Editing	Agent 3 reviews drafts. If rejected, a Correction Note is sent back to Agent 2 and the cycle counter increments.
6	Campaign Assembly Dashboard	All approved content surfaces in the split-panel view alongside the Live Agent Feed. User reviews outputs.
7	Chatbot Corrections	User requests targeted changes via the chatbot. System applies revisions while preserving Fact-Sheet compliance.
8	Export / History	User exports the Campaign Kit as a ZIP file. Project is automatically saved to the History Tracker.

7. Functional Requirements Summary

ID	Requirement	Feature Area	Priority	Status
FR-01	Extract product features and target audience from source document	Agent 1 — Researcher	P0	Planned
FR-02	Generate structured Fact-Sheet in JSON or Markdown	Agent 1 — Researcher	P0	Planned
FR-03	Flag ambiguous statements in source material	Agent 1 — Researcher	P1	Planned
FR-04	Generate 500-word Blog Post from Fact-Sheet	Agent 2 — Copywriter	P0	Planned

ID	Requirement	Feature Area	Priority	Status
FR-05	Generate 5-post Social Media Thread from Fact-Sheet	Agent 2 — Copywriter	P0	Planned
FR-06	Generate 1-paragraph Email Teaser from Fact-Sheet	Agent 2 — Copywriter	P0	Planned
FR-07	Enforce tone differentiation across content formats	Agent 2 — Copywriter	P1	Planned
FR-08	Perform claim-by-claim hallucination check against Fact-Sheet	Agent 3 — Editor	P0	Planned
FR-09	Issue structured Correction Notes on draft rejection	Agent 3 — Editor	P0	Planned
FR-10	Enforce maximum three revision cycles per content piece	Agent 3 — Editor	P1	Planned
FR-11	Display real-time Live Agent Activity Feed	Dashboard	P0	Planned
FR-12	Provide side-by-side source and output comparison view	Dashboard	P1	Planned
FR-13	One-click Campaign Kit ZIP export	Dashboard	P1	Planned
FR-14	Responsive mobile/desktop preview toggle	Dashboard	P2	Planned
FR-15	Secure login, sign-up, and role-based access	Authentication	P0	Planned
FR-16	Natural language AI correction chatbot per content piece	Chatbot	P1	Planned
FR-17	Persistent project history with search and restore	History Tracker	P1	Planned

8. Non-Functional Requirements

Category	Requirement
Performance	Full pipeline execution (Research → Copywrite → Edit → Approve) must complete within 90 seconds for a source document up to 5,000 words.
Reliability	API uptime SLA of 99.5% monthly. Automatic retry on agent failure with exponential backoff up to three attempts.

Category	Requirement
Security	All source documents and generated content encrypted at rest (AES-256) and in transit (TLS 1.3). No user data used for AI model training.
Scalability	System must support concurrent pipeline execution for up to 50 simultaneous users without degradation.
Accessibility	Dashboard must conform to WCAG 2.1 AA standards for keyboard navigation, color contrast, and screen reader support.
Compliance	Data residency configurable per enterprise tenant. GDPR-compliant data deletion within 30 days of user request.

9. Success Metrics & KPIs

Metric	Baseline (Current)	Target (6 Months)
Time to produce 3-channel campaign	4–6 hours (manual)	Under 5 minutes
Brand tone consistency score	Unmeasured	90%+ pass rate on tone audit
Hallucination rate in final content	N/A	Less than 1% of approved outputs
Agent revision cycles per campaign	N/A	Average 1.2 cycles or fewer
User adoption rate	N/A	80% of marketing team active within 60 days
Campaigns exported per week	3–5 (manual)	20+ per week

10. Assumptions & Constraints

Assumptions

- Source documents will be provided in supported formats: PDF, DOCX, TXT, or publicly accessible URLs.
- The underlying LLM provider (e.g., Anthropic Claude or OpenAI GPT-4) will maintain API availability and current capability levels.
- Users possess at least basic digital literacy and do not require onboarding beyond a guided first-run tutorial.

- Brand voice guidelines will be provided by the Marketing team in a structured format during onboarding.

Constraints

- The system must not store raw source documents beyond 90 days without explicit user consent.
- All AI-generated content must remain within the scope of the source document; the agents are not permitted to conduct external web research during content generation.
- The chatbot may not override Agent 3's hallucination flags without explicit user confirmation.
- Maximum source document size is limited to 10 MB or 10,000 words, whichever is smaller.

11. Open Questions

1. What LLM provider(s) should power the three agents — a single provider for consistency, or specialized models per agent role?
2. Should the platform support additional output channels (e.g., press releases, YouTube video scripts) in v1, or are Blog, Social, and Email sufficient for the initial release?
3. What constitutes the definitive brand voice guidelines document, and who in the organization owns its maintenance?
4. Should the History Tracker include performance analytics (e.g., engagement rates from published posts) in a future release via third-party integrations?
5. Is multi-language content generation (i.e., producing outputs in multiple languages from a single English source) a requirement for v1 or a future roadmap item?

12. Revision History

Version	Date	Summary of Changes	Author
1.0	April 2026	Initial draft. Covers all four core features, Authentication, Chatbot, and History Tracker.	Marketing Tech Team

Document Sign-Off